How DNSGuard-X Works

The workflow, step by step, in plain language

DNS is the Internet's phone book: it turns names like `www.google.com` into the numbers (IP addresses) computers use to talk to each other. Because this traffic is almost never blocked, attackers abuse it to steal data, control malware, and overwhelm servers. DNSGuard-X watches these lookups, measures how normal each one looks, and raises a clear warning when something matches a known attack. This page follows a single lookup from start to finish.

The journey of one DNS lookup

1. **Capture — grab the traffic**

   A DNS packet arrives. DNSGuard-X either reads it from a saved recording (a `.pcap` file, safe and repeatable) or sniffs it live from a network card. Live sniffing needs administrator rights.

2. **Parse — pull out the useful facts**

   The packet is decoded into a tidy record: who asked, the name they looked up, the type of record, whether the answer worked, the message sizes, and the time. Anything that isn't DNS is ignored.

3. **Find the real domain**

   The name is tidied and its main registered domain is extracted, so that `a.evil.com` and `b.evil.com` are both recognised as belonging to `evil.com`.

4. **Whitelist — skip the known-safe sites**

   If the domain is a well-known, popular site (one of about 400 in the safe list), the lookup is skipped. This clears away most ordinary traffic so the system can focus on the suspicious remainder — and keeps false alarms low.

5. **Measure the clues**

   DNSGuard-X measures a handful of simple clues about the name (how long it is, how *random* it looks, how many digits and vowels, the record type) and about the computer's behaviour over the last 60 seconds (how many lookups per second, how many different made-up names it tried, and how often the answer was 'no such name'). The most important clue is **randomness**: real names like `google` are predictable and score low, while scrambled or machine-made names score high.

6. **Run the detectors**

   Five independent rules check those clues. A single lookup can trip more than one of them — that is normal and gives stronger evidence. The rules are summarised in the table below.

7. **Raise and tidy the warnings**

   Every rule that fires becomes a warning with a severity and a plain-English reason containing the real numbers (for example, "name is 87 characters and looks scrambled; type TXT"). Repeated warnings for the same computer, domain and type are merged into one, with a count — so a flood of 2,000 shows as a single line that says 'x2000' rather than flooding the screen.

8. **Store and show**

   All warnings and traffic statistics are saved to a small database and displayed: as a summary in the command line, or in the Streamlit and web dashboards with charts, top hosts, and a

filterable table.

The five detectors, in a nutshell

Rule	Plain-English meaning	Severity
D1 – Tunneling (payload)	A single name is very long and looks scrambled — a sign of data hidden inside it.	High
D2 – Tunneling (volume)	One computer looks up many different made-up names under the same domain within a minute.	High
D3 – DGA / water torture	Most of a computer's recent lookups were for names that do not exist — malware hunting for its control server.	Medium
D4 – Flood / Amplification	Too many lookups per second, or a small 'ANY' question that produces a huge answer.	High
D5 – Suspected DGA name	A single name looks machine-generated (very random, few vowels). On its own, only a weak hint.	Low
CP – Cache poisoning	Two different answers come back for the same question, or an answer nobody asked for — a sign of forged replies.	Medium

The whole thing in one line

Capture → read the packet → find the real domain → skip safe sites → measure the clues → run the five detectors → raise and group warnings → save and display.

Everything above is wired together in one function — `DetectionEngine.process_record()` in `dnsguard/engine.py`. If you read only one piece of code to understand the project, read that one.